FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
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ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE(Advanced)-2016**

ANSWERS, HINTS & SOLUTIONS
HALF COURCE TEST –VI
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**A**	**A**	**C**
2.	**A**	**B**	**A**
3.	**D**	**C**	**B**
4.	**A**	**A**	**D**
5.	**A, C**	**A, C, D**	**A, B**
6.	**A, C**	**B, D**	**B, D**
7.	**A, B, C, D**	**A, C, D**	**A, D**
8.	**A, D**	**A, B, D**	**B, C**
9.	**C**	**A**	**A**
10.	**A**	**B**	**B**
11.	**B**	**C**	**A**
1.	**(A) → (p, q, r, s)** **(B) → (p, q, r, s, t)** **(C) → (q, r)** **(D) → (q, r, s, t)**	**(A) → (r, t)** **(B) → (p, q, r, s, t)** **(C) → (p, s, t)** **(D) → (p, q, s, t)**	**(A) → (s)** **(B) → (r)** **(C) → (p)** **(D) → (q)**
2.	**(A) → (q, r, t)** **(B) → (q, r, t)** **(C) → (p, s)** **(D) → (q, t)**	**(A) → (q, t)** **(B) → (s, t)** **(C) → (p, t)** **(D) → (p, t)**	**(A) → (r)** **(B) → (q)** **(C) → (r)** **(D) → (s)**
1.	**2**	**4**	**2**
2.	**9**	**4**	**6**
3.	**6**	**4**	**8**
4.	**3**	**9**	**3**
5.	**2**	**4**	**5**
6.	**5**	**4**	**9**

Physics

PART – I

SECTION – A

1. $\tan\theta = \frac{v^2}{Rg}$ and $\frac{h}{d} = \tan\theta$

3. Static friction depends on tendency.

4. $v_A = \sqrt{\frac{\frac{\lambda L}{4}g}{\lambda}}$

$v_B = \sqrt{\frac{\frac{3L}{4}g}{\lambda}}$

$v_B = \sqrt{3}v_A$

5. Gravitational field inside the cavity is $\vec{E} = \frac{4}{3}\pi\rho G\vec{r}$

where ρ is mass density and $\vec{r}$ is separation between centre of sphere and centre of cavity.
Applying law of Conservation of energy :

$$\frac{1}{2}mV_{esc}^2 - \frac{GMm}{R} + \frac{GMm}{45R} = 0$$

Solving $V_{esc} = \sqrt{\frac{88GM}{45R}}$.

6. $\lambda \propto \frac{1}{T} \Rightarrow T_1 > T_2 > T_3$ as $\lambda_1 < \lambda_2 < \lambda_3$ (Wien's law)

Now as the area under the curve E_λ and λ gives the intensity; so

$\sigma Ae_3T_3^4 > \sigma Ae_2T_2^4$ {Areas of the bodies are same, given}

Now as $T_3 < T_2 \Rightarrow e_3 > e_2$.

7. $P = V^3$, for ideal gas
$PV = nRT$
(A) relation between V and T
$V \times (3V^3) = nRT$
$\Rightarrow V^4 = \left(\frac{nR}{3}\right)T \Rightarrow V^4 \propto T$
(B) Relation between P and T
$PV = nRT \Rightarrow P\left(\frac{P}{3}\right)^{1/3} = nRT$
$\Rightarrow P^{4/3} \propto T$
(C) For expansion
$V \rightarrow$ increases
work-done : positive
internal energy : increases
Hence, heat will have to supplied to the gas.
(D) As $T \propto V^4$
with increase in temperature, volume increases
Hence work done is positive.

8. Both (A) and (D) depend on g.

SECTION – B

1. Adiabatic dQ = 0
Isochoric dW = 0
Isothermal dT = 0 $\Rightarrow$ dU = 0

2. **(A) $\rightarrow$ (q, r, t), (B) $\rightarrow$ (q, r, t), (C) $\rightarrow$ (p, s), (D) $\rightarrow$ (q, t)**

SECTION – C

1. $v = \sqrt{\frac{T}{\mu}}$

T can be calculated by using Hooke's Law and on stretching μ also changes.

2. $f_1 = \left(\frac{340}{340-34}\right)f = \frac{10}{9}f$

and $f_2 = \left(\frac{340}{340-17}\right)f = \frac{20}{19}f$

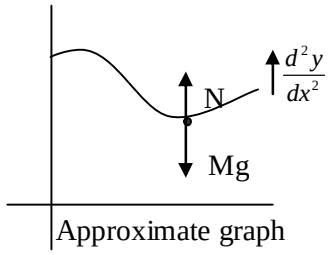
and $\frac{f_1}{f_2} = \frac{\frac{10}{9}}{\frac{20}{19}} = \frac{19}{18}$

3. For 1st reading of oscillator
$f_A = (514 \pm 2)$Hz
$\Rightarrow$ f_A = 516 Hz or 512 Hz
For 2nd reading of oscillator
$f_A = (510 \pm 6)$ Hz
$\Rightarrow$ f_A = 516 Hz or 504 Hz
$\Rightarrow$ A has a frequency of 516 Hz

4. $y = \frac{x^3}{30} - \frac{x^2}{10}$

$\frac{dy}{dx} = \frac{x^2}{10} - \frac{x}{5}$ and at $x = 2; \frac{dy}{dx} = 0$

$\frac{d^2y}{dx^2} = \frac{x}{5} - \frac{1}{5}$ and at $x = 2; \frac{d^2y}{dx^2} = +ve$



Approximate graph

Hence the particle is at it's lowest (minimum) position, in it's path at x = 2 m

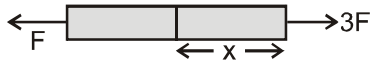
$\frac{d^2y}{dt^2} = 2x\frac{dx}{dt} - 2\frac{dx}{dt}$

$\left(\frac{d^2y}{dt^2}\right)_{at\ x=2} = 20 m/s^2$

$\Rightarrow N - mg = m\frac{d^2y}{dt^2} \Rightarrow N = 1(10+20) = 30N$

$P = f.v = (\mu N)V = \frac{1}{100} \times 30 \times 10 = 3$ watt

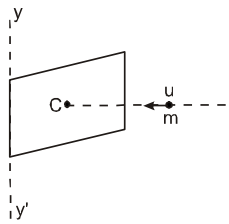
5. Tension in rod at a distance x from right edge is



$T = F(3 - 2\frac{x}{L})$

$\therefore$ net extension in rod $= \int_0^L \frac{T}{YA}dx = \frac{2F}{YA}L$

6. The plate is free to rotate about vertical axis yy'.

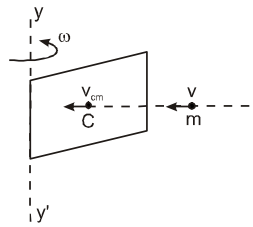


Let v, v_{cm} and ω be the velocity of particle, velocity of centre of mass of plate and angular velocity of plate just after collision.

$\therefore$ From conservation of angular momentum about vertical axis passing though O is

$\therefore \quad mu\ \frac{a}{2} = mv\ \frac{a}{2} + \frac{ma^2}{3}\ \omega$ ....(1)

since the collision is elastic, the equation of coefficient of restitution is



$e = \frac{v_{cm} - v}{u} = 1$ ....(2)

But $v_{cm} = \frac{a\omega}{2}$ ....(3)

solving equation (1), (2) and (3) we get $\omega = \frac{12}{7}\frac{u}{a}$

Chemistry

PART – II

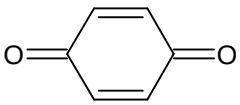
SECTION – A

1. O_2^- has one unpaired electron. It is paramagnetic.

2. In, SO_3^{2-}, S has 3 bond pairs and one lone pair and hence, it is pyramidal in shape not tetrahedral.
Hence (B) is correct.

3. (A). Three possible spin quantum number means an orbital can have at the most three electrons.
So 4th period involves,
3s,3d and 4p subshells.
So, maximum of electrons (and hence elements) = $4s^3 3d^{15} 4p^9 = 3 + 15 + 9 = 27$
(B). All elements of VIII group (i.e. 8, 9, 10 groups) are called typical transition metals. Pt also belongs to 10th group. So it is a typical transition metal.
(C). E.C. of Gd (64) = $[Xe]4f^7 5d^1 6s^2$
So number of f – electrons = 7
(D). Due to ineffective shielding by 3d – electrons in Ga, size of Ga is smaller (almost equal) to that of Al.
Hence **(C)** is correct.

4. $\alpha = \frac{D-d}{(n-1)d} \Rightarrow 0.5 = \frac{40-d}{(1.5-1)d} \Rightarrow 0.5 \times 0.5 \times d = 40 - d$
$\therefore$ 1.25d = 40 $\Rightarrow$ d = 32

5. 
can not exhibit keto enol tautomerism.

6. $Be_2C + 4H_2O \longrightarrow 2Be(OH)_2 + CH_4$
$Al_4C_3 + H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$

7. At larger molar volume, a real gas may behave ideally only to a limit but not always. Hence (B) is incorrect and others (A, C and D) are correct statements.

8. SiO_2 reacts with HF to form H_2SiF_6 is
$SiO_2 + 6HF \longrightarrow H_2SiF_6 + 2H_2O$
Hence A, B, D is correct.

9. Both P & T represents (2R, 3R) – 2, 3 – dibromobutane.
Hence (A) is correct.

10. (I) $\Rightarrow$ Identical & rest of II, III & IV are enantiomers.
Hence (B) is correct.

11. Both Q & S are meso – 2, 3 - dibromobutane.
While (R) is (2S, 3S) - 2, 3 – dibromobutane & hence optically active i.e. lacks in the plane of symmetry.
Hence (C) is correct.

SECTION – C

1. $A = {}^{14}CO_2$
   $C = CO_2$
   $F = H_2$

2. Weight of Fe in 1 mole of haemoglobin = $98600 \times \frac{0.25}{100} = 224$

   number of $Fe = \frac{224}{56} = 4$

3. $\sqrt{\frac{3RT_x}{M_x}} = \sqrt{\frac{2RT_y}{M_y}}$

   $\frac{3 \times 400}{40} = \frac{2 \times 60}{M_y}, M_y = 4$

4. $K_{sp} = 4s^3$

   $S = 3\sqrt{\frac{5 \times 10^{-16}}{4}} = 5 \times 10^{-6}$

   $\left[OH^-\right] = 2s = 2 \times 5 \times 10^{-6} = 10^{-5}, pH = 9$

5. $P_{NH_3} = 2 \times P_{H_2S}$

   $P = 1\,bar, P_{NH_3} = 2\,bar, P_{H_2S} = 1\,bar$

   $K_p = (2)^2 \times 1 = 4$

6. 5th exited state $\Rightarrow n_2 = 6$
   Visible radiation means $\Rightarrow n_2 = 2$
   $6 \rightarrow 2, 5 \rightarrow 2, 4 \rightarrow 2, 3 \rightarrow 2$

Mathematics

PART – III

SECTION – A

1. $a_1, a_2, a_3 \ldots$ are in G.P.

$a_1^{\frac{1}{x_1}} = (a_1 r)^{\frac{1}{x_2}} = (a_1 r^2)^{\frac{1}{x_3}} = \ldots = k$

$\frac{1}{x_1} = \log_{a_1} k$, $\frac{1}{x_2} = \log_{a_2} k$ and $\frac{1}{x_3} = \log_{a_3} k$

Clearly $x_1, x_2, x_3 \ldots$ are in H.P.

2. Equation of tangent will be $\frac{2x}{\sqrt{3}a} - \frac{y}{\sqrt{3}b} = 1$ hence x-intercept is $\frac{a\sqrt{3}}{2}$ and y intercept is $b\sqrt{3}$

$\Rightarrow \quad \frac{1}{2} \times \frac{a\sqrt{3}}{2} \times b\sqrt{3} = 3a^2 \Rightarrow \frac{b}{a} = 4 \Rightarrow e^2 = 17.$

3. For each of the three equations to have equal roots

$4a^2 = 4b$, $4b^2 = 4c$ and $4c^2 = 4a$

$\Rightarrow \quad a^2b^2c^2 = abc \Rightarrow abc = 1$

hence a, b, c can't be integers.

4. We know that in ΔABC, $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

$\therefore$ $\sin A \sin B \sin C = 1 - \cos A \cos B$

$\sin C = \frac{1 - \cos A \cos B}{\sin A \sin B} \le 1$

$\cos(A - B) \ge 1$ i.e., A = B

Hence $\angle C = \frac{\pi}{2}$, $\angle A = \angle B = \frac{\pi}{4}$.

5. Slope of PV = $\frac{2t - 0}{t^2 - 0} = \frac{2}{t}$

$\therefore$ equation of QV is $y = -\frac{t}{2}x$

solving it with $y^2 = 4x$, $Q = \left(\frac{16}{t^2}, \frac{-8}{t}\right)$

Now, as (ΔPVQ) $\frac{1}{2} \times PV \times VQ = 20$ (given)

$PV^2 \times VQ^2 = 40^2$

6. If n = m then equation becomes $z^n = \bar{z}^m$ and it has infinite solution because any $z \in$ real will satisfy it.

If $n \ne m$, let n > m, then $z^n = \bar{z}^m \Rightarrow |z|^n = |z|^m$

$\Rightarrow \quad |z|^m (|z|^{n-m} - 1) = 0$

$\Rightarrow \quad |z| = 0$ or $|z| = 1$, $|z| = 0 \Rightarrow z = 0 + 0i$

$|z| = 1 \Rightarrow z = e^{i\theta} \Rightarrow e^{(m+n)\theta I} = 1$

$\Rightarrow \quad z = 1^{\frac{1}{m+n}}$

$\Rightarrow$ number of solution is m + n + 1.

7. Let $R = \left(3\sqrt{3}+5\right)^9 = [R] + f$ and $\left(3\sqrt{3}-5\right)^9 = f'$, then [R] + f – f′ = an even integer,
which implies f – f′ = 0, Hence [R] is an even number
$[R] = \left(3\sqrt{3}-5\right)^9 - \left(3\sqrt{3}+5\right)^9$ = a multiple of 5
Hence [R] is divisible by 10.

8. If n lines divide a plane is L(n) planes, then
L(n) = L(n – 1) + n
Hence L(10) = 2 + 2 + 3 + … + 10 = 56.

9. For roots of opposite sign a(a – 3) < 0, hence 0 <a < 3

10. Discriminate = $(2a-1)^2 - 4a(a-3) = 8a + 1$
for $a = \frac{n^2+n}{2}$ $D = (2n+1)^2$
Hence roots must be rational numbers.

11. As per data of previous questions the roots will be
$\frac{n^2+3n}{n^2+n}, \frac{n^2-n-2}{n^2+n}$
Hence both the roots lie in [–1, 2].

SECTION – B

1. (A) Equation of any curve through the points in which given lines meet coordinate axes will be $(a_1x - y + b)(a_2x - y + b) + \lambda xy = 0$. If these points are concyclic then $a_1a_2 = 1$.

(B) Any point on $x^2 + y^2 = a_1^2$ will be $(a_1 \cos\theta, b_1 \sin\theta)$.
The chord of contact of the tangents drawn to $x^2 + y^2 = 4$ from this point will be $a_1x \cos\theta + a_1y \sin\theta = 4$. If this touches the circle $x^2 + y^2 = a_2^2$, then $a_1a_2 = 4$.

(C) If the two circles cut orthogonally, then
$-a_1 \times a_1 - a_2 \times a_2 = 2(-1-1) \Rightarrow a_1^2 + a_2^2 = 4$.

(D) Centre of the given circle is (2, –3) and radius is 5. Now distance between (8, 5) and the centre is 10.
Hence $a_1 = 5$ and $a_2 = 15$.

2. (A) $\cos^7 x + \sin^4 x = 1$
$\cos^7 x = (1 + \sin^2 x)\cos^2 x$
$\cos x = 0$ or $\cos^5 x = 1 + \sin^2 x$
$\cos x = 0 \Rightarrow x = \frac{-\pi}{2}, \frac{\pi}{2}$, $\cos^5 x = 1 + \sin^2 x$
$\Rightarrow x = 0$.
$\therefore\ x = \frac{-\pi}{2}, 0, \frac{\pi}{2}$.

(B) $\sqrt{3}\,\text{cosec}\,20^\circ - \sec 20^\circ$
$= \frac{\sqrt{3}}{\sin 20^\circ} - \frac{1}{\cos 20^\circ} = \frac{\sqrt{3}\cos 20^\circ - \sin 20^\circ}{\sin 20^\circ \cos 20^\circ}$
$= \frac{2\left[\frac{\sqrt{3}}{2}\cos 20^\circ - \frac{1}{2}\sin 20^\circ\right]}{\sin 20^\circ \cos 20^\circ} = \frac{4\cos 50^\circ}{\sin 40^\circ} = 4$

(C) $4\cos 36^\circ - 4\cos 72^\circ + 4\cos 18^\circ \cos 36^\circ$

$= 4\left(\frac{55+1}{4}\right) - 4\left(\frac{55-1}{4}\right) + 4\left(\frac{55-1}{4}\right)\left(\frac{55+1}{4}\right)$

$= 3$

(D) $\operatorname{cosec} x = 1 + \cot x$

$\frac{1}{\sin x} = \frac{\sin x + \cos x}{\sin x} \Rightarrow \cos x + \sin x = 1$ and $\sin x \neq 0$

$\cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$x - \frac{\pi}{4} = -2\pi + \frac{\pi}{4}$ or $x - \frac{\pi}{4} = \frac{\pi}{4}$, $\quad x = \frac{-3\pi}{2}, \frac{\pi}{2}$.

SECTION – C

1. Let these lines be given by $y = mx$ for $m = m_1, m_2 \& m_3$
Then substituting $y = mx$ in the given equation gives
$m^3 - cm^2 - bm + 1 = 0$
Now $m_1 + m_2 + m_3 = b$, $m_1m_2 + m_2m_3 + m_3m_1 = -c$
$m_1m_2m_3 = -1$ & $m_1m_2 = -1$
Hence $b + c = 2$

2. $C_0C_1 + C_1C_2 + \ldots + C_{n-1}C_n = {}^{2n}C_{n-1} = \frac{2n!}{(n-1)!(n+1)!}$

$\Rightarrow \quad S_{n+1} = \frac{(2n+2)!}{n!(n+2)!} \Rightarrow \frac{S_{n+1}}{S_n} = \frac{4n^2+6n+2}{n^2+2n} = \frac{15}{4}$

$\Rightarrow \quad n = 2$ or 4.

3. $P(2, 4) = (2t^2, 4t) \Rightarrow t = 1$

$t' = -t - \frac{2}{t} \Rightarrow t' = -3 \Rightarrow R$ is $(18, -12)$

Also $tt' = 2$ gives $t' = 2$ hence $Q(8, 8)$
Perpendicular bisector of PR : $x - y = 14$ and of PQ : $3x + 2y = 27$
solving together gives $h = 11, k = -3$
$h + k = 8$

4. $N = 34x5y$
number is divisible by 4 if 5y is divisible by 4 i.e. $y = 2, 6$
if $y = 2$ then $3 + 4 + x + 5 + 2$ must be divisible by $9 \Rightarrow x = 4$
if $y = 6$ then $3 + 4 + x + 5 + 6$ must be divisible by $9 \Rightarrow x = 0, 9$.

5. Note that the two given lines are mutually perpendicular, hence their point of intersection i.e. (1, 1) will lie on director circle of the ellipse.
If centre of the ellipse is (h, k) the equation of direction circle will be $(x - h)^2 + (y - k)^2 = 25$
as (1, 1) lies on this circle hence locus of center will be
$(h - 1)^2 + (k - 1)^2 = 25$
$\therefore$ radius of circle is 25.

6. We have $a = \frac{2\Delta}{h_A}, b = \frac{2\Delta}{h_B}, c = \frac{2\Delta}{h_C}$

$\Rightarrow \quad a = \Delta, b = \Delta, c = \frac{2\Delta}{3}$

$\Rightarrow \quad a + b + c = 2\Delta\left(\frac{1}{h_A}+\frac{1}{h_B}+\frac{1}{h_C}\right)$

$2s = 2\Delta\left(\frac{1}{2}+\frac{1}{2}+\frac{1}{3}\right) = 2\Delta\left(\frac{8}{6}\right)$

$\Rightarrow \quad s = \frac{4}{3}\Delta$

Now $\Delta^2 = s(s-a)(s-b)(s-c)$

$\therefore \quad \Delta^2 = \frac{4}{3}\Delta\left(\frac{\Delta}{3}\right)\left(\frac{\Delta}{3}\right)\left(\frac{2\Delta}{3}\right)$

$\Rightarrow \quad 1 = \frac{8}{81}\Delta^2 \Rightarrow 8\Delta^2 = 81$

$\Rightarrow \quad 2\sqrt{2}\Delta = 9.$